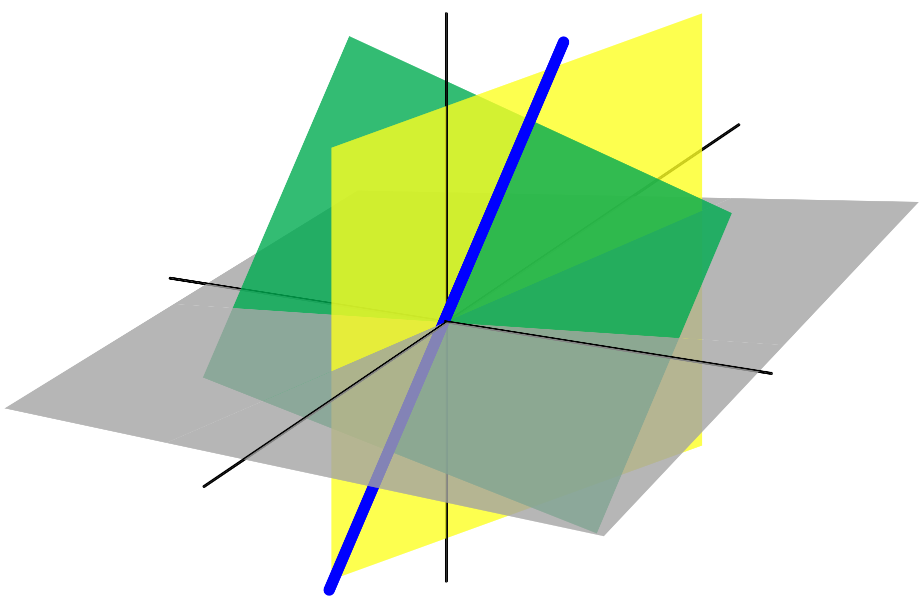
Mathematical Methods in Signal Processing

TAMU ECEN-601

Brody Jordan

Fall 2025



Acknowledgements

These notes were compiled from the course materials for ECEN-601 (Mathematical Methods in Signal Processing) taught by Dr. Krishna Narayanan at Texas A&M University during the Fall 2025 semester.

Mathematical Methods in Signal Processing (ECEN-601)

Instructor:	Dr. Krishna Narayanan
Semester:	Fall 2025
Time:	TTh 2:20pm - 3:35pm
Location:	ETB 1003

Description: Representations and algorithms for signal processing; linear algebra, vector spaces, normed and inner product spaces; projection, orthogonalization, rank-nullity theorem; matrix representations, singular value decomposition; sampling, Fourier analysis, spectral methods; statistical signal processing and linear estimation.

Contents

Chapter 1

Introduction

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\begin{array}{ccccccc}
\forall x, \forall y, P(x,y) & \Rightarrow & \exists x, \forall y, P(x,y) & \Rightarrow & \forall y, \exists x, P(x,y) & \Rightarrow & \exists y, \exists x, P(x,y) \\
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\forall y, \forall x, P(x,y) & \Rightarrow & \exists y, \forall x, P(x,y) & \Rightarrow & \forall x, \exists y, P(x,y) & \Rightarrow & \exists x, \exists y, P(x,y)
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